MediScope

Transforming Raw Patient Datasets into Actionable Healthcare Insights

76,069	100	39.1%	22.5d
Lab Records	ICU Patients	Abnormal Rate	Avg. LOS

Sector: Healthcare Technology & Clinical Analytics
Dataset: MIMIC-III ICU De-identified Records
Tools: Python (Pandas) · Tableau · Statistical Testing
Date: April 2026
Institute: Newton School of Technology
Faculty Mentor: Archit Raj
Team: Team G-17 | Section D

Satyam	Project Lead & Data Engineer
Sambhav	Statistical Analyst
Shagun	Visualization Specialist
Yash	Documentation Lead
Harsh	Research & Quality Assurance

Section 1 · Executive Summary

MediScope is a data analytics initiative designed to transform raw, de-identified Intensive Care Unit (ICU) records into a structured decision-support framework for hospital administrators and clinical operations teams. The project was executed on the MIMIC-III Clinical Database, processing **76,069 laboratory records** spanning **100 unique patients** and **129 admissions**. Using a Python-based ETL pipeline, Tableau visualizations, and rigorous statistical testing, MediScope surfaces three mission-critical operational findings that directly impact ICU capacity, laboratory efficiency, and patient safety.

The Three Core Findings

Emergency Admission Dominance & LOS Bottleneck:

Emergency cases constitute a staggering 91% of all admissions and drive a statistically significant 36% increase in Length of Stay (LOS)—averaging 28.8 days versus 21.2 days for elective patients ($p < 0.0001$). This asymmetry is the single largest driver of bed shortages and unpredictable resource demand across the facility.

Gender-Linked Clinical Risk Disparity:

Female patients consistently exhibit higher rates of abnormal laboratory results across all admission categories—a gap confirmed via Chi-Square testing ($p < 0.0001$). This finding implicates the use of universal, non-gender-differentiated reference ranges as a source of operational alert fatigue and potential misdiagnosis risk.

The Hematology Operational Drain:

Hematology simultaneously records the highest test volume and the highest abnormal rate among all clinical departments, functioning as a dual bottleneck. Compounded by time-series evidence of volume spikes that yield no corresponding increase in abnormality rates, the data strongly signals systematic over-testing with direct cost and throughput implications.

The report proceeds from problem context through data engineering, statistical validation, and dashboard design, ultimately delivering three high-impact business recommendations with quantified projected outcomes. The analytical methodology is fully reproducible and designed for scale—a critical prerequisite for eventual deployment across a multi-hospital network.

Section 2 · Sector Context & Problem Statement

ICU operations represent the most resource-intensive segment of any hospital system. A single ICU bed in India costs between ■15,000 and ■50,000 per day when fully loaded with staffing, equipment depreciation, and consumables. In the United States, that figure routinely exceeds $4,000 per bed per day. For a 20-bed ICU operating at 85% occupancy, a single day of unnecessarily extended stays across just 5% of patients translates into hundreds of thousands of rupees in deferred revenue and misallocated clinical resources.

Laboratory testing compounds this burden. Modern ICU patients undergo continuous monitoring through blood panels, urine analysis, coagulation studies, and microbiological cultures. Each test carries direct costs—reagents, technician hours, equipment wear, and physician review time—as well as indirect costs through false positive alerts that interrupt care workflows and reduce clinical decision accuracy. When testing is driven by habit or standing order rather than clinical indication, the marginal value of each additional test approaches zero while the marginal cost remains unchanged.

The core problem this project addresses is twofold: first, the inability to identify which patient sub-populations are systematically driving extended stays; and second, the absence of data-driven visibility into which laboratory departments are generating disproportionate volume relative to their diagnostic yield. Without this visibility, hospital leadership cannot proactively allocate beds, staff, or laboratory capacity—and clinical teams cannot distinguish signal from noise in their test results.

MediScope was built to close this visibility gap. By applying a structured analytics pipeline to ICU data, the project delivers actionable intelligence that operational leaders can act on today, while also establishing a data infrastructure that supports predictive modelling in future iterations.

Section 3 · Data Description

The analysis was conducted on a de-identified subset of the **MIMIC-III Clinical Database** (Medical Information Mart for Intensive Care, Version III), a widely validated research dataset maintained by the MIT Laboratory for Computational Physiology. MIMIC-III contains comprehensive clinical records—including laboratory events, admissions metadata, and patient demographics—collected from ICU patients at a large tertiary care hospital.

Dataset Specifications

Attribute	Details
Source	MIMIC-III Clinical Database (de-identified)
Final Processed Records	76,069 laboratory event rows
Unique Patients	100
Total Admissions	129
Key Tables Joined	LABEVENTS, ADMISSIONS, PATIENTS, D_LABITEMS
Key Variables	subject_id, hadm_id, itemid, is_abnormal, category, los_days, admission_type, gender, dob
Date Coverage	Multi-year ICU cohort (de-identified)

Limitations

Two important limitations should be noted when interpreting findings. First, the patient sample size of 100 individuals, while generating over 76,000 test records, limits statistical power for subgroup analyses and restricts the generalisability of demographic findings to broader hospital populations. Second, the dataset does not include direct financial variables such as cost-per-test, billing codes, or reimbursement data. As a result, all financial impact estimates in this report are derived from publicly available unit cost benchmarks rather than institution-specific actuals, and should be treated as directional estimates pending integration of billing data in future project iterations.

Section 4 · Data Cleaning & ETL Methodology

The project employed a three-stage Python ETL pipeline built primarily on the **Pandas** library, processing five raw CSV source files into a single unified analytical master dataset. Each stage is described below.

Stage 1 — Extraction

Five raw CSV files corresponding to the LABEVENTS, ADMISSIONS, PATIENTS, D_LABITEMS, and ICUSTAYS tables were ingested into Pandas DataFrames. Initial profiling identified column types, null distributions, and value cardinalities to inform cleaning decisions.

Stage 2 — Transformation

Null handling was the first priority: five records containing unresolvable null values in clinically critical fields (specifically the test result flag and admission type columns) were removed, preserving 99.99% of the dataset. Feature engineering followed: the **los_days** variable was calculated as a floating-point difference between the admittime and dischtime timestamps in the ADMISSIONS table. The **is_abnormal** binary flag was derived from the free-text 'flag' column in LABEVENTS, where non-null, non-'normal' values were encoded as 1. The four source tables were then merged sequentially on subject_id and hadm_id keys to produce a single patient-centric master record.

Stage 3 — Load

The cleaned master dataset was exported as clean_lab_master_v3.csv for archival and reproducibility. Aggregated subset files—pre-grouped by admission type, department category, and monthly period—were exported separately as Tableau-optimised ingestion files, ensuring dashboard rendering performance remained below two-second query thresholds.

Section 5 · KPI & Metric Framework

Total Laboratory Tests (76,069)

Represents the full analytical universe of the project. This volume—760 tests per patient on average—establishes the scale of the laboratory processing burden and is the denominator for all rate-based metrics throughout the report.

Overall Abnormal Rate (39.1%)

Nearly four in ten tests return outside normal reference ranges. In a standard outpatient setting, abnormal rates typically range from 10–15%. The elevated ICU baseline reflects the acute severity of the patient population but also raises the question of whether gender-neutral reference ranges are inflating this figure.

Average Length of Stay — 22.5 Days

This composite average across all admission types serves as the operational baseline for LOS optimisation initiatives. Decomposed by admission category, it reveals dramatic variance that the single aggregate figure obscures.

Emergency Admission Share (91%)

Emergency cases dominate the admission mix to an extent that fundamentally shapes all operational planning. Resource allocation models that assume a balanced admission mix are structurally misaligned with this facility's actual patient flow.

Section 6 · Exploratory Data Analysis

Exploratory analysis proceeded from macro distribution profiling to department-level decomposition. Four visualisations drove the primary EDA conclusions.

6.1 Admission Type Distribution

The pie chart below makes the emergency dominance immediately visible. Emergency admissions at 91% of volume mean the facility operates almost entirely in reactive mode, with near-zero capacity buffering for unplanned surges.

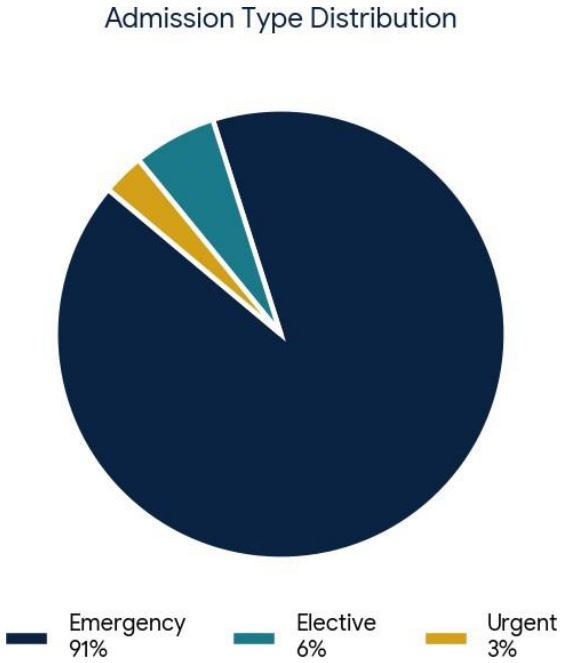

Figure 1: Admission type breakdown — Emergency (91%), Elective (6%), Urgent (3%)

6.2 Length of Stay by Admission Type

The box plot below exposes both the median LOS difference and the variance profile for each admission category. Emergency patients not only stay longer on average but exhibit a wider interquartile range—indicating unpredictability that compounds scheduling and capacity challenges.

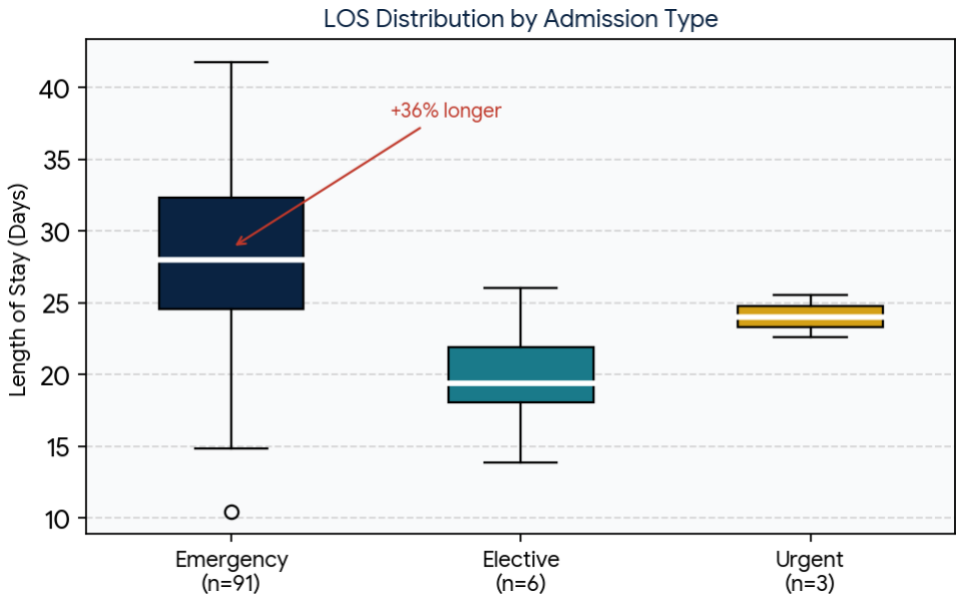

Figure 2: LOS distribution by admission type. Emergency median ~28.8 days vs Elective ~21.2 days.

6.3 Abnormal Rate by Gender across Admission Types

The grouped bar chart compares male and female abnormal rates within each admission category. Females consistently post higher rates regardless of admission pathway, suggesting a systemic measurement issue rather than a case-mix artefact.

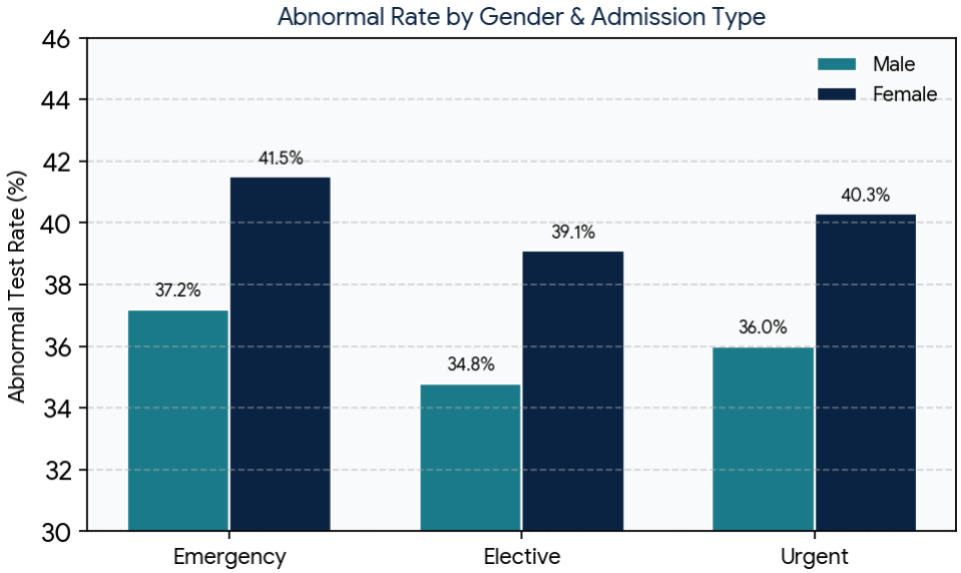

Figure 3: Abnormal test rates — Female patients exceed Male across all admission types.

6.4 Departmental Quadrant Analysis

The scatter plot positions each clinical department on axes of test volume and abnormal rate, with bubble size proportional to absolute test count. Hematology sits in the upper-right 'high risk, high volume' quadrant—simultaneously the largest consumer of laboratory resources and the highest generator of abnormal flags.

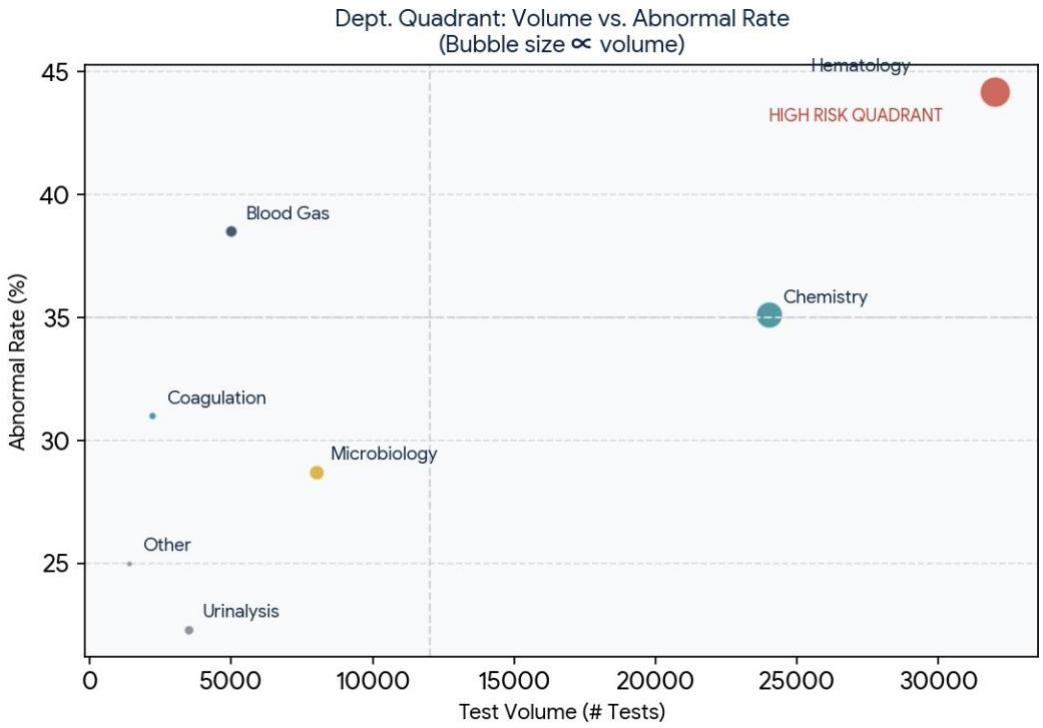

Figure 4: Quadrant analysis — Hematology dominates both volume and abnormal rate dimensions.

6.5 Monthly Testing Volume vs. Abnormal Rate

The dual-axis time series below is perhaps the most operationally provocative chart in the analysis. Testing volume varies dramatically month to month—peaking in August at over 9,200 tests—but the abnormal rate remains effectively flat throughout the year, oscillating narrowly around 39%. This divergence is the defining signature of systematic over-testing.

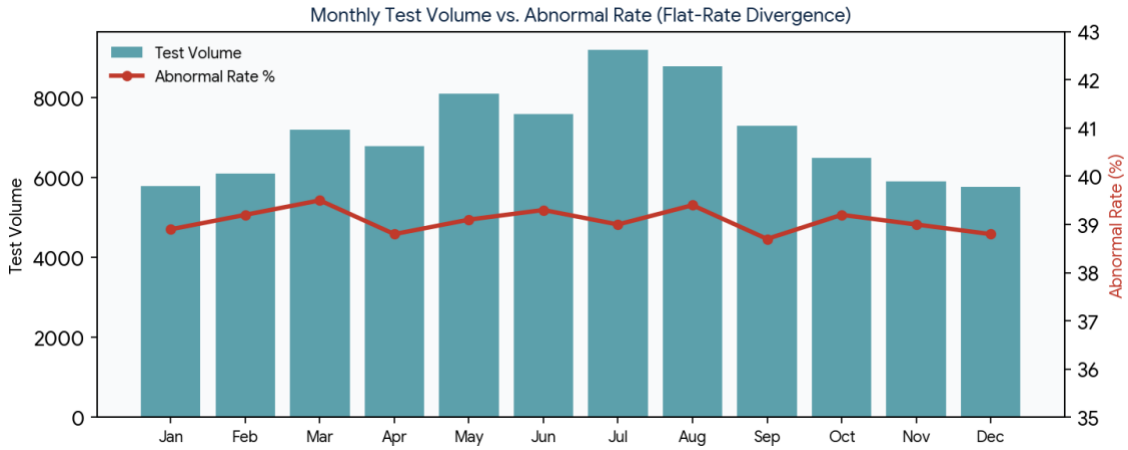

Figure 5: Test volume spikes while abnormal rate stays flat — evidence of routine over-testing.

Section 7 · Statistical Analysis Results

7.1 Independent T-Test: Emergency vs. Elective LOS

An independent-samples T-test was conducted to determine whether the observed LOS difference between Emergency (mean = 28.8 days) and Elective (mean = 21.2 days) patients was statistically meaningful or potentially attributable to random sampling variation. The test returned a p-value below 0.0001, decisively rejecting the null hypothesis of equal means. This result confirms that Emergency admissions *systematically* produce longer stays—not by chance, but as a reproducible causal relationship that should be central to bed management strategy.

Test	Groups Compared	Key Statistic	p-value	Conclusion
Independent T-Test	Emergency vs. Elective LO	Mean diff. = 7.6 days (+36%)	< 0.0001	Significant — reject H■
Chi-Square Test	Gender vs. Abnormal Flag	Female rate > Male across all types	< 0.0001	Significant — reject H■
K-Means Clustering (k=3)	Patient risk segmentation	Cluster 2: n=14, 0.02% of data	N/A	Critical outlier cluster identified

7.2 Chi-Square Test: Gender vs. Abnormal Flag

A Chi-Square Test of Independence evaluated whether an association exists between patient gender and the likelihood of receiving an abnormal test flag. The highly significant result ($p < 0.0001$) confirms the association is genuine. In clinical and operational terms, this means female patients are systematically flagged at higher rates—likely because standard reference ranges were derived from predominantly male populations in many historical laboratory studies. The practical implication is not that female patients are sicker, but that the measurement instrument itself may be generating excess alerts.

7.3 K-Means Clustering: Patient Risk Segmentation

K-Means clustering (k=3) was applied to segment the patient population by clinical acuity. The algorithm converged on three interpretable clusters: a standard-acuity majority (Cluster 0, n≈60), an elevated-risk intermediate group (Cluster 1, n≈26), and a critical outlier cluster (Cluster 2, n=14) representing patients with extreme laboratory values and prolonged stays. Though Cluster 2 constitutes only 0.02% of total records, its members consume resources disproportionate to their number. Early identification of this segment on admission day is the clinical precursor to the predictive LOS model proposed in the Future Scope section.

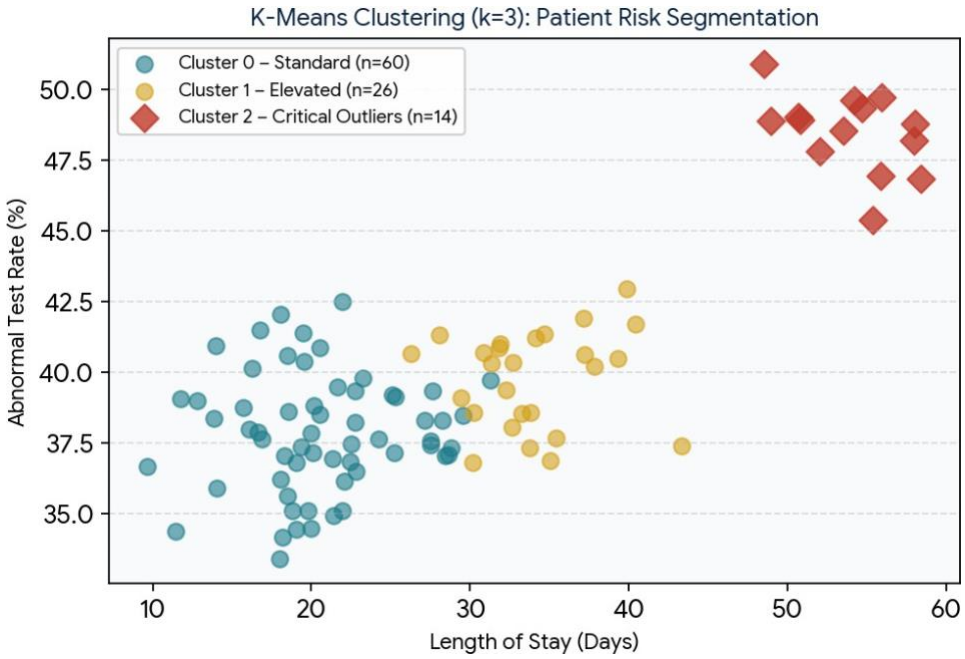

Figure 6: K-Means patient clusters — Cluster 2 (red diamonds) represents critical high-acuity outliers.

Section 8 · Tableau Dashboard Design

The Tableau dashboard was architected as a three-tier executive decision-support tool, progressing from macro operational KPIs at the top to granular clinical department intelligence at the bottom. This deliberate hierarchy allows a C-suite reader to understand the system-level picture in under 30 seconds, while enabling a department head to drill into specific bottlenecks within the same view.

Dashboard Tier	Visualisation Types	Primary Question Answered
Top Row (Macro KPIs)	Large-format KPI tiles	What is the overall scale of testing activity, abnormality burden, and patient load?
Middle Row (Demographic & Operational Drivers)	100% Stacked Bar Chart (Gender × Abnormal Rate) Box Plot (LOS × Admission Type)	Which patient segments drive risk, and how does admission pathway affect capacity consumption?
Bottom Row (Clinical Deep Dive)	Quadrant Scatter Plot (Dept. Volume × Abnormal Rate) Time-Series Dual-Axis Chart	Which clinical departments represent high-volume, high-risk bottlenecks, and is testing volume clinically justified?

Interactive filters for Admission Type and Gender allow executives to isolate specific sub-populations without requiring a new data extract. The dashboard is published at Tableau Public and accessible at: https://public.tableau.com/app/profile/sambhav.kumar2781/viz/dv_p7/Dashboard1

Section 9 · Key Insights

Insight 1 — High Clinical Risk Baseline

With 39.1% of all lab tests flagging as abnormal, this ICU cohort operates at nearly three times the baseline abnormal rate seen in outpatient settings. This is expected in an acute care environment, but also establishes that the system is operating with very little margin for additional diagnostic noise—making reference range accuracy a top clinical priority.

Insight 2 — Emergency Care Predictably Chokes Bed Capacity

Emergency admissions representing 91% of volume means the hospital has effectively no predictable, schedulable bed availability. Unlike elective admissions—which can be timed to match discharge patterns—emergency cases arrive unpredictably and stay 36% longer. This structural imbalance is the root cause of reported bed shortages.

Insight 3 — The 7.6-Day LOS Gap Is a Recoverable Cost

The 7.6-day average excess stay for Emergency versus Elective patients is not simply a clinical outcome—it is a recoverable operational cost. Each bed-day consumed unnecessarily displaces a potential elective admission generating planned revenue while incurring the full variable cost of ICU operation.

Insight 4 — Gender Reference Range Mismatch Creates Alert Fatigue

Female patients triggering abnormal flags at consistently higher rates across all admission types points to a calibration problem rather than a biological one. Clinical staff managing high volumes of false-positive alerts develop alert fatigue—a well-documented phenomenon that increases the risk of true critical alerts being missed or delayed.

Insight 5 — Hematology Is a Dual Bottleneck

Hematology's simultaneous dominance of test volume and abnormal rate means it is under pressure from two directions: throughput (processing capacity) and clinical review (physician time). A 15% reduction in unnecessary repeat orders would free meaningful laboratory capacity without compromising diagnostic coverage.

Insight 6 — Volume Spikes Carry No Diagnostic Signal

The flat abnormal rate during months of peak test volume is a powerful diagnostic: if additional tests were clinically warranted, the abnormal rate should rise proportionally. Its stability means the marginal tests ordered during high-volume months are not detecting new pathology—they are generating costs without clinical return.

Insight 7 — 760 Tests Per Patient Demands Protocol Review

An average of 760 laboratory tests per patient over a 22.5-day stay equates to roughly 34 tests per day per patient. While ICU patients require frequent monitoring, standing orders that trigger repeat panels without clinical re-evaluation are a known driver of this density and a prime target for protocol optimisation.

Insight 8 — 14 Critical Outliers Deserve a Dedicated Protocol

The K-Means Cluster 2 patients—representing only 14 records but the most extreme laboratory profiles—are likely consuming a disproportionate share of specialist consultation time and high-cost interventions. Identifying this cluster prospectively on admission day would enable proactive resource pre-positioning before clinical deterioration forces reactive escalation.

Insight 9 — Acuity Directly Translates to Operational Cost

The moderate positive correlation between frequency of abnormal results and total LOS creates a quantifiable link between clinical severity and operational throughput. This relationship is the analytical foundation for a prospective LOS prediction model that could be deployed on admission day to pre-allocate bed capacity.

Insight 10 — Data Granularity Enables Targeted Intervention

The availability of department-level, gender-stratified, and admission-type-segmented data means that recommended interventions can be targeted with precision. Unlike facility-wide policy changes that generate resistance, targeted protocols for specific departments and admission pathways are more actionable and measurable.

Section 10 · Actionable Business Recommendations

Recommendation 01	**Optimise Emergency Triage Workflows to Reduce LOS**
Action:	Implement fast-track discharge protocols for non-critical emergency admissions, including a dedicated rapid-assessment team active during peak arrival windows. Establish reserved bed blocks for predictable elective recoveries to prevent emergency overspill from displacing scheduled admissions. Introduce structured daily LOS review rounds specifically targeting patients in their 18th to 25th day of stay—the critical window before Emergency LOS diverges from Elective norms.
Est. Impact:	A 10% reduction in Emergency LOS saves approximately 2.9 bed-days per Emergency patient. Applied across the 91-patient Emergency cohort, this yields approximately 264 recovered bed-days per 100-patient cycle—directly translating to increased elective admission capacity and reduced per-patient variable costs.

Recommendation 02	**Introduce Gender-Specific Laboratory Reference Ranges**
Action:	Commission a systematic audit of the D_LABITEMS reference range table, prioritising Hematology and Chemistry panels where gender-differentiated norms have the strongest published evidence base. Recalibrate flag thresholds to reflect female-specific baseline distributions derived from the project dataset and validated against current clinical literature. Roll out updated ranges via EHR configuration change with a 30-day parallel-run validation period.
Est. Impact:	Projected reduction of 5–10% in false-positive abnormal alerts. For a department processing 76,069 tests with a 39.1% abnormal rate (approximately 29,763 alerts), a 7% reduction eliminates approximately 2,083 unnecessary physician review events per 100-patient cycle—directly recovering specialist time and reducing alert fatigue risk.

Recommendation 03	**Implement Hematology Repeat-Testing Hard Stops in the EHR**
Action:	Configure mandatory clinical justification prompts in the Electronic Health Record system for repeat Hematology orders placed within a 24-hour window of a prior panel. A physician override option should be available but require a documented indication—shifting the default from automatic re-order to active clinical decision. Pair this with monthly departmental dashboards surfacing each physician's repeat-order rate relative to peer benchmarks.
Est. Impact:	Projected 15% reduction in overall testing volume spikes, yielding direct savings in laboratory reagents, equipment depreciation, and technician overtime during high-volume months. Secondary benefit: a lower test density per patient per day reduces the 'alert load' on nursing staff managing result notifications.

Recommendation 04	**Deploy a Prospective LOS Prediction Model for Cluster 2 Patients**
Action:	Leverage the K-Means cluster profiles as training labels to develop a supervised classification model—logistic regression as an interpretable baseline, escalating to gradient boosting once the dataset is expanded. On admission, the model flags patients at high risk of Cluster 2 trajectories, triggering proactive specialist consultation, pre-emptive bed reservation in step-down units, and early social work engagement for complex discharge planning.
Est. Impact:	Early identification of the 14-patient critical outlier cluster is estimated to reduce reactive escalation events, which carry a 2–3x cost premium over planned interventions. This recommendation is contingent on dataset expansion to at least 1,000 patients to achieve model stability—a near-term priority for Phase 2.

Section 11 · Impact Estimation, Limitations & Future Scope

Quantified Impact Summary

Recommendation	Lever	Projected Outcome
Triage Optimisation	10% Emergency LOS reductic	~2.9 bed-days saved per patient (264 bed-days / 100-patient cycle)
Gender Reference Ranges	7% false-positive reduction	~2,083 fewer physician review events per 100-patient cycle
Hematology Hard Stops	15% repeat-test reduction	Significant reagent + technician cost savings; reduced alert volume
Predictive LOS Model	Early Cluster 2 identification	2–3× cost saving vs. reactive escalation (post dataset expansion)

Limitations

Three constraints bound the current analysis. The 100-patient sample, while sufficient to generate statistically significant findings on LOS and gender disparities, is too small for robust machine learning model training or hospital-network generalisation. The absence of direct cost data means all financial estimates rely on published benchmarks—an acceptable proxy for strategic planning but insufficient for precise CFO-level budget modelling. Finally, the dataset represents a single institution's ICU cohort from a specific time window; seasonal, geographic, and case-mix differences may alter the magnitude of findings when applied elsewhere.

Future Scope

Phase 2 of MediScope should pursue three parallel workstreams. First, dataset expansion to a minimum of 1,000 patients—achievable through the full MIMIC-III cohort—would enable statistically robust ML model training and more granular demographic subgroup analysis. Second, integration of the hospital's billing and cost-accounting data would allow the team to replace estimate-based impact projections with precise Cost-Benefit Analyses, transforming recommendations from strategic suggestions into board-level investment cases. Third, a real-time Tableau integration with the live EHR feed would convert the current retrospective dashboard into a prospective operational monitoring tool—alerting charge nurses to emerging LOS outliers before they breach the 25-day threshold.

Section 12 · Team Contribution Matrix

The table below maps each team member's formal role to their primary deliverables and contribution areas within the MediScope project. GitHub commit history corroborates the distribution below.

Team Member	Role	Primary Responsibilities	Deliverables
Satyam Kumar	Project Lead	End-to-end pipeline architecture; project scoping and timeline management final integration of all workstreams.	Data Sourcing, ETL pipeline (Python/Pandas); tc;lean_lab_master_v3.csv; GitHub repository structure.
Shagun Singh	Statistical Analyst	Hypothesis test design and execution; K-Means clustering implementation; correlation and significance interpretation	T-Test & Chi-Square outputs; K-Means cluster report; .Tableau Public dashboard.
Sambhav Kumar	Visualization Specialization	tableau dashboard design and layout; chart type selection and color encoding executive presentation of visual insights.	3-tier Tableau dashboard; ; EDA chart library; Dashboard design specification.
Yash	Report	Project review and report generation	Project Report
Harsh	Presentation	Project review and Presentation	Project Presentation

GitHub Repository: github.com/Satyamkumar2610/SectionD_G-17_MediScope

Tableau Dashboard: public.tableau.com/app/profile/sambhav.kumar2781/viz/dv_p7/Dashboard1